

DOUVALUE KNOWLEDGE CENTER · PORT HARCOURT, RIVERS STATE, NIGERIA



FARM OPERATIONS SCHEDULE

REVISION 5 · WEEK-BY-WEEK MANAGER'S GUIDE

Greenhouse Bell Pepper | Open-Field Habanero | Rukpokwu Extension | Prevention System & Diagnosis Kit

BUILT ON PREVENTION

Every task sits inside the Root-Cause Prevention System. Five gates make the four Season-1 failures unable to recur — clean soil, live scouting, diagnosis before treatment.

WHAT'S NEW IN REV 5

Prevention gates woven through; a clean-restart protocol for replanting Greenhouse 4 & 5; the Rukpokwu ridge establishment plus its own full week-by-week schedule; a complete pest & disease diagnosis kit; day-numbered weeks; and the farm logo.

HOW TO USE IT

Count weeks from your transplant date. Follow each day's steps in order. GO = proceed; HOLD = stop and fix first. Record every action, with a photo, in the logbook.

Sources: Haifa Crop Guides · Rutgers NJAES · FAO Hot Pepper Production · UC IPM · ISHS · Port Harcourt climate data | Prepared July 2026

HOW TO USE THIS GUIDE

Read this page once before you start — then follow it week by week.

Counting your weeks

- Day 1 = the day you transplant into the ground. That week is **Week 0**.
- Week 1 = days 2–8; Week 2 = days 9–15; and so on.
- Formula: $(\text{today} - \text{transplant date}) \div 7 = \text{week number}$.
- Write the transplant date in the logbook — every week counts from it.

Reading each row

- **Day** — which day to do the task. **Task** — the activity type.
- **What to do** — step by step, in order; do not skip steps.
- **Watch for** — the warning sign to look for while doing it.
- **GO / HOLD** — green = proceed; red = stop and fix the problem first.

Your greenhouse (passive net-wall ventilation)

- Polythene roof + UV-net side walls. Ventilation is **passive** through the nets — there are no operable roof vents.
- Nets lower humidity 5–10%, but insects enter as freely as outdoors.
- Install yellow sticky traps and scout every Day 2; act on pests immediately.
- Open doors 10 AM–2 PM daily to flush the mid-day humidity the nets can't clear alone.

The logbook rule

- Record every action: date, activity, product, dose, IRAC/FRAC group, observations.
- Attach a photo — it is the evidence a task really happened.
- The logbook protects you if a crop fails and enables early diagnosis.
- In the app, each of these becomes a scouting log, spray log or diagnosis entry.

THE PREVENTION SYSTEM — WHY REVISION 5 EXISTS

Season 1 was lost to four causes. Every task in this schedule now sits inside a structure built to make those causes unable to repeat.

Revision 5 is not just a new calendar — it is the calendar wired to a **Root-Cause Prevention System**. The rule behind it is simple: **prevention must be structural, not personal**. A system that only works while everyone stays diligent fails the moment one person doesn't. So each cause is blocked by a **gate** the season cannot pass without documented evidence — a clean soil report, a scouting log, a diagnosis. When those are impossible to skip, last season's post-mortem does not get rewritten next year.

The four root causes of the Season 1 loss

RC1 · Nematode soil, never tested	RC2 · Thrips controlled too late	RC3 · Blind remote oversight	RC4 · Diagnose-by-guess
Plants entered the season with galled roots that cannot take up water or nutrients. No spray fixes a galled root — they were doomed at day one.	Thrips carry tospovirus. By the time spraying began the virus was already systemic — the remedy killed the insect after the damage was done.	A days-long problem took weeks to reach the owner. No scouting, no trigger, no ground truth between rotations.	The team sprayed what was visible (thrips) without establishing what was killing the plants. You cannot fix a cause you never named.

The five gates — a season cannot move forward until each is cleared

GATE	WHEN	BLOCKS	PASS CONDITION — THE EVIDENCE THAT CLEARS IT
0 · Ground Clearance	Before planting	RC1	Soil + nematode lab report returned and cleared by the agronomist. If positive: solarise / remediate / re-site and re-test until clear. No seedling goes in until this is green.
1 · Establishment Readiness	Before transplant	RC2	Full input list on-site incl. thrips controls; drip pressure-tested; spacing pegged and photographed; sticky traps installed; SOPs live; scout roster set. Evidence: signed Establishment Checklist.
2 · Standing Controls	Continuous	RC2 · RC3	Daily scouting against trap thresholds; escalation triggers live; owner verifying daily/weekly. Evidence: daily logs, dashboard, escalation register.
3 · Diagnosis before Treatment	Any symptom	RC4	No treatment applied until the cause is diagnosed. Symptom → diagnostic protocol → confirmed cause → matched remedy. Evidence: a diagnosis entry that precedes any spray entry.
4 · Cycle Close & Learn	End of cycle	Drift	Root inspection of sample plants; control audit; SOPs updated from what was learned. Evidence: Cycle Review document.

The recurrence test — run it every cycle

CAUSE	PROOF IT IS WORKING	RED FLAG IT IS FAILING
RC1	A cleared lab report is on file before any transplant.	Planting started with no soil report.
RC2	Daily trap counts logged; threshold-cross to spray under 24 h.	Trap counts gapped >2 days, or thrips reach "heavy" with no prior alert.

CAUSE	PROOF IT IS WORKING	RED FLAG IT IS FAILING
RC3	Scouting compliance >95%; dashboard shows fresh data daily.	The owner learns of a problem from a dead plant, not from an alert.
RC4	Every treatment has a diagnosis recorded before it.	A spray is logged with no diagnosis preceding it.

Where you see a **GATE** tag on a week below, that task is the point where a gate is cleared or held. The full control layer lives in the Root-Cause Prevention System document; this schedule is how it runs day to day.

SECTION A — GREENHOUSE BELL PEPPER

5 greenhouses · yellow & red varieties · polythene roof + passive UV-net walls · drip irrigation. Gate 0 (soil) and Gate 1 (readiness) clear at Week 0; Gate 2 runs every day after.

WEEK 0 · Day 1–7 Soil Preparation, Pre-Plant Treatments & Transplant ESTABLISHMENT

Gates 0 + 1

▲ **UV-net side walls = passive ventilation (no operable roof vents).** Airflow lowers humidity 5–10% vs a sealed house — a real disease advantage — but whitefly, thrips and aphids enter as freely as outdoors. Yellow sticky traps and the Day 2 scouting round are not optional.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
3–4 WKS BEFORE	SOIL SOLARISATION + NEEM CAKE	- Cover all beds with CLEAR (transparent) polythene, pulled tight, edges sealed. Leave 3–4 weeks — PH sun drives soil to 55–70°C, killing nematodes, fungal spores, cutworms and weed seed. - Never black plastic (it insulates, does not heat). No air pockets. - At 2 weeks before transplant: 250 g neem cake per planting hole, mixed in. - Never Furadan (carbofuran) — banned, destroys soil biology. Neem cake + solarisation is safer and works longer. - End of period: lift plastic, air beds 2 days, then transplant week.	Clear plastic only, laid flat, no air gaps.	GO Clear plastic; 3–4 wks; neem cake at Wk-2. HOLD Black plastic or under 3 wks — fix / extend.
Day 1	SOIL FINAL PREP + EC/pH	- Urea 50 g/hole (3 tbsp). Compost 20–30 kg/bed into top 10 cm (yearly target 5 t/ha). - EC/pH every week from now. pH 5.8–6.5 (if >7.0 add citric acid 5 g/10 L, re-test). Plain-water EC 1.8–2.8 mS/cm (>3.5 = flush 60 min). - Run drip 60 min to settle beds and verify every emitter; mark and fix blocked ones.	Water not draining within 10 min = blockage. Never transplant into flooded beds.	GO Draining well; emitters flowing; urea applied. HOLD Waterlogged or >5 blocked emitters — fix first.
Day 3 · 4–6 PM	PRE-PLANT FUNGICIDE	- Mancozeb 80WP 2.5 g/L (40 g / 16 L). Spray beds, floor, posts and the inside of the net walls. 4–6 PM only. Mask + gloves. - Log 'Mancozeb, FRAC M3' — this opens your FRAC rotation.	Wind-driven rain through the nets dilutes it; beds should look faintly dusty 2 h later.	GO Beds dry; 4–6 PM; calm air. HOLD Rain-laden wind through nets — wait.
Day 5 · 4–7 PM	PRE-PLANT INSECTICIDE	- Cypermethrin 10EC 1 ml/L (16 ml / 16 L). Spray all interior surfaces — net walls both faces, floor, posts, under benches. - Do this within 48 h of transplant. Doors shut overnight. Rinse knapsack 3×. - Log 'Cypermethrin, IRAC 3A' — the next insecticide MUST be a different group.	Net walls let insects recolonise fast — don't spray a week early and leave doors open.	GO 4–7 PM; doors shut; logged 3A. HOLD Strong wind/rain — reschedule.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 6 · 4–6 PM	TRANSPLANT + VENTILATION PROTOCOL	- Transplant 4–6 PM only. 1 seedling/hole at 30 cm, firm gently, water each row as you plant. - Ventilation (daily, whole cycle): open all doors 10 AM–2 PM to flush the mid-day humidity spike the passive net walls can't clear alone; close at 2 PM to hold evening warmth. - Install yellow sticky traps: 1 per 6 m² at plant height. - Drip 30 min after planting. Logbook: today = WEEK 0, DAY 1 — every future week counts from here.	Seedlings must be hardened 7+ days; soft lush ones collapse.	GO Overcast PM; hardened; soil pre-moist; traps up. HOLD Mid-day heat; unhardened; dry soil.
Day 7	WATER + DAY-1 CHECK	- Drip 6:30 AM 20 min · 1 PM 15 min (skip only if net walls show condensation AND soil moist 5 cm deep) · 5 PM 20 min. - Walk every house. Day 1–3 mid-day wilt with evening recovery = normal transplant shock. - Still collapsed by 6 PM from Day 3 on: pull one and read the roots — white firm = water only; brown slimy = rot; red/brown galls = nematode.	A persistent wilter's roots tell you the cause. Read them, don't guess.	GO Mild mid-day wilt, evening recovery. HOLD Full wilt unrecovered by 6 PM Day 3 — investigate roots now.

WEEK 1 · Day 8–14 Root Establishment — Biostimulant Begins, Pest Monitoring Active ESTABLISHMENT

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	MICROBIAL INOCULANT DRENCH	- Trichoderma harzianum 5 g/L (mix in 50 L, into 1,000 L tank, drip 30 min) — every 14 days, whole cycle. Or Bacillus subtilis per label / Azospirillum 10 ml/L. - These replace yeast (yeast breeds fungus-gnats and imbalance in humid soil). - NEVER within 48 h of any fungicide drench — fungicide kills the organisms you just paid for.	Inoculant Day 1 → fungicide no earlier than Day 3. Never the same tank.	GO Biweekly; 48 h clear of fungicide; soil moist. HOLD Within 48 h of fungicide — wasted.
Day 2	IRRIGATION + IPM SCOUTING + TRAP CHECK	- Every Day 2, all weeks: inspect 10 plants/house (50 total) — leaf undersides, growing tips, stem base. Log pest, count, location. - Thresholds — whitefly >5/trap or >2/leaf; thrips >10/trap or >3/tip → advance to Spinosad (IRAC 5); aphids on >5% of plants → spray. - Count every sticky trap and record it. A rising trend week-on-week means advance the spray — don't wait. Add Bt if caterpillar eggs/larvae appear.	Rising trap counts = a population building. This is the early warning Season 1 never had.	GO <5 whitefly/trap — hold the schedule. HOLD >5 whitefly/trap → Thiamethoxam (4A) tomorrow.
Day 3	GARLIC–CHILLI SPRAY	- Steep 6 g garlic + 6 g chilli in 4 L hot water 6+ h. STRAIN TWICE. Add 15 ml soap + 15 ml vinegar → 15 L. - Spray 4–7 PM: leaf undersides + stem bases. PHI 0 days. - Postpone 1 h if leaves are wet.	Wet leaves dilute it; particles block the nozzle.	GO Leaves dry; strained clean; 4–7 PM. HOLD Leaves wet — wait.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 4	IRRIGATION + PLANT ASSESSMENT	- By end of Week 1 every plant should be upright with new tip growth. - Pale NEW tip leaves = nitrogen short (raise NPK Day 6). Yellow LOWER leaves = over-watering (cut PM drip 5 min). - Pull any persistent wilter and read the roots — problems found now are treatable; found at Week 4 they are not.	Catch root trouble early — that is the whole game.	GO Upright, dark green, new tip growth. HOLD Pale tips = add NPK; yellow lower = less water.
Day 6	INSECTICIDE (IRAC 4A) + NPK DRENCH	- Thiamethoxam 25WG (IRAC 4A) 0.2 g/L (3.2 g / 16 L). Spray 4–7 PM, undersides + tips. Log 'IRAC 4A'. - NPK 20-10-10: 1 kg in 50 L → 1,000 L tank → drip 25 min. - Rotate IRAC every spray: 3A → 4A → 5 → 28 → repeat. Check the logbook before mixing.	Same IRAC group twice = resistance. Verify last group first.	GO 4A confirmed different from last; logged. HOLD Do not repeat 3A this week.
Day 7	WATER + RECORD	- Drip as standard (never skip Day 7 water). - Write the end-of-week summary: average height, observations, inputs used. - Source bamboo/iron stakes (120 cm) now — needed Day 1 Week 3.	Water is 7 days a week, no exceptions.	GO Water as normal; stakes on order. HOLD —

WEEK 2 · Day 15–21 Vegetative Growth — Nutrition, Neem, Ca+K Begin VEGETATIVE

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	Ca + K FOLIAR	- Ca Nitrate 50 g + K Nitrate 50 g / 16 L. Spray 4–7 PM, all surfaces incl. undersides. - Foliar now because small roots absorb slowly — leaf uptake is 8–10× faster at this stage. - NEVER mix Ca Nitrate with Mg Sulphate — they cancel each other out.	Tip scorch after spray = sprayed in heat. Wait for cooler air after 4 PM.	GO 4–7 PM; leaves cool; no rain through nets. HOLD Mid-day heat; wet leaves.
Day 2	NEEM OIL + MICROBIAL INOCULANT	- Inoculant (biweekly) if not already applied this week — not within 48 h of fungicide. - Neem: 150 ml COLD-PRESSED neem + 30 ml soap, stir, add 1 L warm water until milky, fill 16 L. Use within 8 h. - Spray 4–7 PM, undersides + stem junctions.	Oil separating = weak emulsion; add 10 ml soap and re-shake.	GO Cold-pressed; fresh mix; after 4 PM. HOLD Heat-extracted neem; mix >8 h old.
Day 3	IRRIGATION + MID-WEEK CHECK	- Target Week 2: 15–25 cm tall, dark green, lateral branches forming at the base. - Prepare stakes (120 cm, 1/plant) — installed Day 1 Week 3. - Check and record sticky-trap counts.	Pale new growth = raise NPK next application.	GO 15–25 cm; dark green; branching. HOLD Pale tips — increase NPK.
Day 4	Mg SULPHATE FOLIAR	- Magnesium Sulphate 50 g / 16 L. Spray 4–7 PM both surfaces. - Mg is the core of chlorophyll; Rivers State coastal soils are chronically Mg-short. - NEVER mix with Ca Nitrate.	Interveinal yellowing on lower leaves = Mg deficiency; go weekly if seen.	GO 4–7 PM; separate from Ca. HOLD Mixed with Ca Nitrate.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 6	FUNGICIDE (FRAC M1) + INSECTICIDE (IRAC 5)	- Copper Oxychloride 50WP (FRAC M1) 3 g/L — foliage + soil surface. Log; rotation now M3 → M1. - Spinosad 45SC (IRAC 5) 0.3 ml/L — AFTER 4 PM only (UV degrades it). Focus tips + undersides (thrips). Log; rotation 3A → 4A → 5. - Never the same FRAC or IRAC group twice running.	Repeat groups create permanent resistance.	GO Different FRAC from last; Spinosad after 4 PM. HOLD Same FRAC/IRAC as previous.
Day 7	WATER + WEEK-2 REVIEW	- End-of-week logbook entry. Confirm stakes ready for Day 1. - Review IRAC/FRAC groups used; plan next rotation.	No branching by Day 14 = raise nitrogen.	GO Stakes ready; branching on most plants. HOLD No branching — increase NPK N.

WEEK 3 · Day 22–28 CRITICAL — Staking, First Pruning, 30 cm Protocol Begins

VEGETATIVE

Gate 2 live

▲ 30 cm spacing pruning starts this week and never stops. Prune every Day 3 and Day 5 for the crop's whole life. The goal is a visible 5 cm air gap between neighbours at all times — merged canopies spread disease in under 48 h in PH humidity. Sanitise the blade between every single plant.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	SPLIT Ca + K DRENCH + STAKING	- Split Ca from now: 0.7 kg Ca Nitrate Day 1 + 0.7 kg Day 4 (steady supply prevents Blossom-End Rot). - Day 1: Ca Nitrate 0.7 kg in 30 L → tank → drip 25 min → flush lines 10 min. Then K Nitrate 1 kg in 50 L → SEPARATE pass. - Ca and K must be separate passes — mixed they form a white precipitate that blocks emitters. - Install all stakes (120 cm, 5 cm from stem, 30 cm deep). Tie loosely — 2 fingers between stem and tie. Record tank EC (1.8–2.8).	White crust on emitters = Ca and K were mixed; flush 60 min at full flow.	GO Ca/K separate passes; stakes in. HOLD Ca + K in one tank — never.
Day 3	PRUNING (CANOPY-TRIGGERED)	- Sanitise shears 30 s in 70% alcohol / 10% bleach BEFORE you start and BETWEEN EVERY PLANT (mosaic virus travels on the blade). - Remove all laterals below the first Y-fork; remove every leaf touching a neighbour; strip yellow/damaged/diseased leaves. - Confirm 5 cm air gap all round. Spray cut wounds with dilute copper (2 g/L). Never remove >30% of a plant's leaves at once.	Tool sanitation between plants is the single most important disease control you have.	GO Blade sanitised each plant; 5 cm gaps. HOLD Skipping sanitation — one cut can lose a row.
Day 4	FUNGICIDE (FRAC 4+M3) + INSECTICIDE (IRAC 28)	- Metalaxyl-M + Mancozeb (FRAC 4+M3) 2.5 g/L — foliage AND soil drench around bases (systemic; kills Phytophthora from inside). Log; rotation M3 → M1 → 4+M3. - Chlorantraniliprole 200SC (IRAC 28) 0.5 ml/L, 4–7 PM. Log. - Never use Metalaxyl-M twice running — resistance forms fast.	Metalaxyl-M is your strongest systemic — use only in rotation.	GO >14 days since last fungicide; both rotated. HOLD Metalaxyl-M consecutively — never.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 5	PRUNING #2 + TIE CHECK	- Second prune, same sanitised protocol. Remove suckers grown since Day 3 (2–3 cm/day in PH heat). - Check every tie — a Day 1-loose tie can cut the stem by Day 5; re-tie to 2 fingers. - Copper on cut wounds. If the 5 cm gap re-merges within 72 h, ease off nitrogen next Day 6.	Skip Day 5 and the canopy merges over the weekend — disease entry.	GO Second prune done; ties checked; gaps held. HOLD Skipping Day 5 prune.
Day 6	NPK 15-15-15 + BORON FOLIAR BEGINS	- NPK 15-15-15: 1 kg in 50 L → tank → drip 25 min. - Boron (Borax) EXACTLY 1 g/L (16 g / 16 L) — tips and young leaves only, never >1.5 g/L. Weekly through fruit set (Wks 3–9). - Boron prevents hollow pods / poor set — an endemic Niger-Delta deficiency. Keep 4 h apart from Ca.	1 g/L works; 2 g/L burns; 3 g/L damages. Weigh it — never eyeball.	GO Borax exactly 1 g/L; tips only. HOLD >1.5 g/L — irreversible scorch.
Day 7	WATER + PRUNING AUDIT	- Walk all houses. Count plants where the Y-fork is clearly visible with a 5 cm gap. - Target ≥90%. Below 80% — raise pruning intensity next Day 3 & Day 5.	Y-fork buried on >20% of plants = behind on pruning; fix by Week 4.	GO >90% with visible Y-fork and gap. HOLD <80% — increase pruning now.

WEEK 4 · Day 29–35 Pre-Flowering — Watch for First Buds, Intensify Nutrition PRE-FLOWERING

Gate 2 live

› First flower bud = three switches. (1) Stop foliar Ca permanently, go drench-only (Ca can't move from leaf to fruit). (2) Begin Zinc foliar. (3) Move to high-K NPK from next Day 6.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	Ca + K DRENCH + PRUNING #1	- Ca Nitrate 1 kg / K Nitrate 1 kg — separate passes, flush between. If buds visible, raise Ca to 1.4 kg. - Full sanitised prune; remove outward-leaning branches that can't be tied upright. - Re-check all ties — budding branches gain weight and slip.	From first OPEN flower, Ca Nitrate is never foliar — it damages pollen.	GO Ca/K separate; pruning done. HOLD First open flower → stop foliar Ca.
Day 3	ZINC FOLIAR + PRUNING #2	- Zinc Sulphate 2 g/L (32 g / 16 L), 4–7 PM — monthly from now. Zn drives capsaicin/pigment and pollen viability. - Full sanitised prune; tie every flower-bearing branch individually.	Grey metallic sheen within 24 h = mild Zn toxicity; drop to 1.5 g/L.	GO Zn 2 g/L; 4–7 PM; pruned. HOLD Grey discolouration — reduce dose.
Day 4	Mg DRENCH + INSECTICIDE (ROTATE)	- Mg Sulphate 1 kg drench, separate pass (not with Ca). - Insecticide: rotate to the next IRAC group vs the logbook (if last was 28 → 3A Lambda-cyhalothrin 0.5 ml/L; if last was 5 → 28). 4–7 PM.	Correct Mg deficiency now — very hard to fix after fruit set.	GO Mg separate pass; IRAC rotated. HOLD Mg + Ca in one tank.
Day 5	PRUNING + GARLIC SPRAY	- Full sanitised prune; clear suckers at/below the Y-fork. - Garlic–chilli (biweekly): 6 g + 6 g steeped, strained twice, 15 ml soap + 15 ml vinegar, 15 L. Spray tips/bud zones 4–7 PM.	Thrips gather in flower buds pre-flowering — hit the tips.	GO Strained clear; tips sprayed; 4–7 PM. HOLD Particles in mix — re-strain.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 6	HIGH-K NPK SWITCH + FUNGICIDE (M3)	- Switch to NPK 15-5-20 (or 15-15-15 + 500 g KNO3/1,000 L) from now to harvest — K drives fill, colour and sugar. - Mancozeb 80WP (FRAC M3) 2.5 g/L — completes rotation M3 → M1 → 4+M3 → M3; restart at M1 next time.	Low K after flowering = pale, underweight fruit and marginal leaf scorch.	GO High-K applied; FRAC M3 logged. HOLD Staying on low-K after buds.
Day 7	WATER + FRUIT-SET PREP	- Count and record plants with visible buds. - Confirm all open-flower plants are on Ca drench, not foliar. From next week raise Ca to 1.4 kg/application.	Brief the whole team: no foliar Ca anywhere once flowers open.	GO Open-flower plants on Ca drench. HOLD Any foliar Ca on open flowers.

WEEK 5–6 · Flowering Full Flowering — Pollination Protection, Ca+K Intensified FLOWERING

Gate 2 + Gate 3

▲ **Never spray open flowers.** Spray only after 5 PM when flowers have partly closed; never in wind through the nets. Copper or any chemical on open anthers ruins pollen and the poor set only shows 7–10 days later — too late to fix.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	Ca + K DRENCH (INCREASED) + PRUNING	- Ca Nitrate 1.4 kg / K Nitrate 1.4 kg per 1,000 L — separate passes. Density at 30 cm needs 30–40% more nutrient per m². - Full sanitised prune; tie all flower-bearing stems; remove leaf-only non-productive stems that block light/air.	>10% flower drop in a day → check irrigation, Ca dose, and heat (>35°C).	GO Increased Ca/K; branches tied. HOLD Flower drop spike — investigate.
Day 3	BORON SPRAY + PRUNING	- Boron 1 g/L on tips only (through Week 9) — never on open flowers. - Prune touching leaves above the Y-fork (upper canopy merges in flowering). - If set <60%, gently vibrate flower stems at noon on clear days to move pollen.	Set <60% within a week of flowering = act within 3 days (pollen damage, heat, or boron).	GO Boron on tips; hand-pollinate if low set. HOLD Boron onto open flowers.
Day 4	FUNGICIDE (M1) + INSECTICIDE (ROTATE)	- Copper Oxychloride (FRAC M1) 3 g/L — foliage + soil, avoid open flowers. - Insecticide: next IRAC group in sequence. - Both sprays AFTER 5 PM only during flowering.	Copper into open flowers 10 AM–3 PM kills pollen within 2 h.	GO After 5 PM; FRAC + IRAC rotated. HOLD Spraying the 10–3 pollination window.
Day 6	HIGH-K NPK + Mg + NEEM	- High-K NPK fertigation; Mg Sulphate 1 kg separate pass; biweekly neem oil (150 ml, fresh emulsion) 4–7 PM. - Check fertigation EC 2.0–3.0 mS/cm; >3.5 flush 1 h then half-dose.	White salt crust on media = overfeeding; flush before fertigating.	GO EC 2.0–3.0; no crust; neem fresh. HOLD EC >3.5 or crust — flush first.

WEEK 7–8 · Fruit set Fruit Set & Early Development — Maintain Calcium Drench FRUIT DEVELOPMENT

Gate 2 + Gate 3

Ca drench is now the most important input — never miss one. Blossom-End Rot shows at Week 9–10 but is caused by a Ca gap NOW. Keep pruning and staking tight; fruit weight leans stems and merges canopies again.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	SPLIT Ca + K DRENCH + STAKE CHECK	- Ca split 0.7 kg Day 1 + 0.7 kg Day 4 (1.4 kg/week). K Nitrate 1.4 kg separate pass. - Walk every row; re-stake any leaning plant same day (leaning fruit = humidity trap = Botrytis). - Prune leaves shading fruit (colour needs light) but keep leaves shading the root zone. EC 2.0–3.0.	A leaning fruit-laden plant traps humidity around the fruit — ideal Botrytis.	GO Upright; Ca/K separate; fruit lit. HOLD Leaning plants — re-stake today.
Day 3	PRUNING + FRUIT CHECK (BER / ANTHRACNOSE)	- Full sanitised prune; 5 cm gaps. - Check fruit bottoms for BER (dark sunken spot) → raise Ca drench to 1.8 kg/1,000 L at once. - Check fruit surface for anthracnose (irregular dark spots) → 7-day fungicide interval + airflow.	Once BER shows on a fruit it's lost; raising Ca now saves the rest.	GO No BER; fruit firm/green; steady water. HOLD BER found — Ca to 1.8 kg now.
Day 5	PRUNING + FUNGICIDE (ROTATE)	- Full sanitised prune. - Fungicide 10–14 day interval (10 days in rainy season) — rotate group, spray 5–7 PM around fruit bases and leaf-on-fruit contact points (Botrytis).	Rainy season (Jun–Sep): humid air through nets lifts internal humidity — go to 10-day interval.	GO FRAC rotated; interval matches season. HOLD 14-day interval in heavy rain.
Day 6	HIGH-K NPK + Mg + INSECT ROTATION	- High-K NPK; Mg Sulphate 1 kg separate pass; insecticide next IRAC group 4–7 PM. - Zinc foliar (monthly) if Week 8. Boron weekly until set complete, then biweekly.	<50% fruit set by Week 8 → review Ca/K/boron and check for spray on open flowers.	GO Fruit on >80% of plants; IRAC rotated. HOLD <50% set — investigate.

WEEK 9–12 · Fruit fill & colour Fruit Fill & Colour — PHI Compliance from Week 10 FRUIT FILL & COLOUR

Gate 2 + Gate 3

PHI compliance from Week 10. Harvest begins Wk 10–12. From Week 10 use PHI-0 organics only — Neem oil, Garlic–chilli, Copper Hydroxide. NO Cypermethrin, Thiamethoxam or Chlorantraniliprole within 14 days of harvest; keep a 21-day buffer for export. Residue failures lose whole consignments.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 1	Ca + K DRENCH (CONTINUE)	- Ca Nitrate 1.4 kg + K Nitrate 1.4 kg — separate passes, every Day 1 and Day 4 to last harvest. - Continue twice-weekly pruning; tie heavy fruit branches individually. - Colour check: red/yellow varieties begin turning Wk 10–11 — photograph progress.	Irregular water now = both BER (deficit) and cracking (sudden excess). Keep it regular.	GO Ca/K 1.4 kg separate; on track for colour. HOLD Cracking — you under-then over-watered.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
Day 3 & 5	PRUNING + ORGANIC SPRAYS ONLY (Wk 10+)	- Maintain airflow around fruit (Botrytis loves the moist zone under big fruit). - From Week 10 organics only: Neem oil 150 ml/16 L (Day 2) alternating Garlic–chilli (Day 4); Copper Hydroxide 2 g/L (PHI 0) after 5 PM. - Stop all synthetics + Metalaxyl-M. Walk houses daily; log first ripe-fruit date.	Switching to organics at Week 10 is both PHI compliance and market quality.	GO Synthetics stopped by Wk 10; organics in use. HOLD Any synthetic within 14 d of harvest.
Day 6	REDUCED NPK + Ca MAINTAINED + HARVEST GRADING	- NPK to 0.7 kg/1,000 L (half) — less N so energy goes to ripening. Ca stays 0.7 kg Day 1 + 0.7 kg Day 4 until the last fruit is off. - Grade every batch: A export/premium (uniform, full colour, firm, 70–90 mm), B local (minor marks), C processing (undersized/irregular, no rot — always sold). - Cut with clean scissors, 1 cm stem, handle gently — bruised fruit can't store or export.	Selling A as B loses 20–30% per kg; grading takes 15 min/batch.	GO Reduced NPK; Ca held; scissors sanitised. HOLD Reducing Ca before last harvest.

SECTION B — OPEN-FIELD HABANERO

~1,000 plants · 30 cm spacing · drip irrigation · full sun. Higher pest pressure than the houses — perimeter scouting and pod-borer monitoring are the difference.

WEEK H-0 · Land prep + transplant Land Preparation, Mulching & Transplant ESTABLISHMENT

Gates 0 + 1

▲ **Open field = no overhead cover.** Rain washes off spray (reapply within 4 h of >15 mm rain); full sun dries soil faster; and bare soil splashes Phytophthora onto crowns — **mulch to 12 cm** is essential at 30 cm spacing. Wind helps airflow, so disease pressure is a little lower than the greenhouse.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
3–4 WKS BEFORE	SOLARISATION + NEEM CAKE + COMPOST	- Clear polythene over the whole field 3–4 weeks (open-field sun reaches 65–70°C — one cycle drops nematodes 70–80% in the top 20 cm). - Neem cake 250 g/hole at 2 weeks before; compost 5 t/ha (2 kg/m²) incorporated before solarisation. No Furadan. - After: lift plastic, air 2 days, land prep. Water pH 5.8–6.5, EC 1.8–2.8.	Transparent plastic only; solarisation is even stronger in open sun.	GO Clear plastic 3–4 wks; neem + compost in. HOLD Black plastic or <3 wks.
Day 1–3	LAND PREP + 12 cm MULCH	- Urea 50 g/hole mixed in. - Mulch (dry grass / rice straw / coco peat) to 12 cm across the ENTIRE field incl. between plants — at 30 cm spacing there is almost no bare soil, cover it all. - Run drip 60 min to settle; verify emitters.	Mulch under 8 cm in heavy rain = Phytophthora crown-rot outbreak.	GO Mulch 12 cm; drip flowing; neem applied. HOLD Mulch below 8 cm — top up.
Day 5 · 4–6 PM	TRANSPLANT HABANERO	- Transplant 4–6 PM only. 30 × 30 cm, each plant over an emitter, straight rows. Water each row as you plant. - Install 100 cm stakes now (habanero gets bushy and fruit-heavy). - Logbook: transplant date = HABANERO WEEK H-0, Day 1.	Habanero resists shock better than bell pepper but still wilts in mid-day sun.	GO 4–6 PM; hardened; stakes ready. HOLD Mid-day transplant under full sun.
Day 6–7	WATERING + DAY-1 CARE	- Field irrigation, all weeks: 6:30 AM 30 min (dry) / 20 min (rainy); 5 PM 20 min (skip if >15 mm rain today). After >25 mm rain, skip next morning entirely. - Walk the field, flag any plant not upright. Day 1–3 wilt normal; not recovered by Day 3 evening → read roots.	Open-field soil dries fast in PH's ~33°C sun; the 6:30 AM run builds a buffer before peak heat.	GO Drip twice daily; plants recovering. HOLD Full drip after >25 mm rain — root suffocation.

WEEK H-1 to H-2 · Establishment Root Establishment — Microbial Inoculant, Ca+K, IPM ESTABLISHMENT

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-1 · Day 1	MICROBIAL INOCULANT (500 L field tank)	- Trichoderma 2.5 g/L (mix in 25 L, into 500 L tank, drip 30 min) — every 14 days. Or Bacillus subtilis / Azospirillum 5 ml/L. - Open-field soil holds Trichoderma better, building a lasting protective rhizosphere vs Phytophthora/Fusarium/nematodes. - Keep 48 h clear of any fungicide drench.	48 h gap is stricter in field — fungicide persists longer in the soil.	GO Biweekly; 48 h clear; soil moist. HOLD Same day as fungicide.
H-1 · Day 3	Ca + K FOLIAR (FIELD)	- Ca Nitrate 50 g + K Nitrate 50 g / 16 L, 4–6 PM, both surfaces (habanero leaves are waxy). - Reapply next dry evening if rain within 2 h. Continue foliar through Week 3, then drench.	Check the sky — sudden PH showers Jun–Oct wash it off.	GO Clear 2 h after spray; 4–6 PM. HOLD Rain within 2 h.
H-1 · Day 6	INSECTICIDE (IRAC 4A) + NPK 20-10-10	- Thiamethoxam (IRAC 4A) 0.2 g/L, 4–7 PM. From Week 6 add Chlorantraniliprole (IRAC 28) for pod borer. - NPK 20-10-10: 1 kg in 500 L field tank → drip 25 min. Log IRAC 4A.	Field has higher thrips/aphid/pod-borer pressure — check perimeter plants first.	GO IRAC 4A rotated; edges checked. HOLD —
H-2 · Day 3	PRUNING #1 (30 cm PROTOCOL)	- Sanitise between every plant. Remove laterals below the Y-fork and any leaf touching a neighbour. Confirm 5 cm gap. Copper 2 g/L on wounds. - Twice weekly (Day 3 + Day 6) from here. Habanero is bushier — expect more volume.	Habanero canopy closes faster than bell pepper at the same spacing.	GO Sanitised; 5 cm gap; wounds treated. HOLD Skipping field pruning.
H-2 · Day 6	FUNGICIDE (M3) + NPK + Mg	- Mancozeb (FRAC M3) 2.5 g/L on dry leaves only (needs 2 dry hours to bind); delay if rain. - NPK 15-15-15 1 kg / 500 L; Mg Sulphate 500 g separate pass. Log FRAC M3.	Mancozeb on wet leaves won't adhere.	GO Dry leaves; no rain 2 h; 4–7 PM. HOLD Rain / wet leaves — wait.

WEEK H-3 to H-5 · Vegetative Vegetative Frame — Stakes, Boron, Pruning Established VEGETATIVE

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-3 · Day 1	STAKING + Ca+K DRENCH SWITCH + BORON	- Stakes 100 cm, 1/plant. Switch Ca+K to drench: Ca Nitrate 500 g → 500 L → 20 min → flush 5 min; K Nitrate 500 g separate pass (field tank = half greenhouse quantity). - Boron foliar begins 1 g/L on tips, weekly through fruit set.	Hollow-pod from boron deficiency is as common in habanero here as in bell pepper.	GO Stakes in; Ca/K separate; Boron 1 g/L. HOLD Ca + K in one pass.
H-3 → H-5 · Day 3 & 6	IRRIGATION + TWICE-WEEKLY PRUNING + SPRAY ROTATION	- Prune twice weekly, sanitised, 5 cm gaps, re-tie after each prune. - H-3 Day 3 garlic / Day 6 NPK + Copper Oxy (M1) + insecticide rotate. H-4 Day 3 neem / Day 6 Ca+K + NPK + Metalaxyl (4+M3) + Spinosad (5). H-5 Day 3 garlic / Day 6 high-K + Boron + Mancozeb (M3). - Zinc 2 g/L monthly (first now).	Check every crown at the soil line for Phytophthora lesion; if found, drench that plant + 4 neighbours with Metalaxyl-M.	GO Twice-weekly prune + rotation logged. HOLD Crown-rot lesion — Metalaxyl-M drench now.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-3 – H-5 · Day 2	HELICOVERPA POD-BORER MONITORING	- Inspect perimeter plants every Day 2 from Week 3. Look for cream/brown caterpillars boring stems/buds, entry holes with dark frass. - Threshold: any 3 plants with active entry → Chlorantraniliprole (IRAC 28) 0.5 ml/L, 4–7 PM. Log 'IRAC 28 — pod-borer threshold'.	Damage inside a pod is invisible until you cut it — monitor weekly, act early.	GO Perimeter checked Day 2; act at 3 plants. HOLD >10 plants with entry holes — spray whole field today.

WEEK H-6 to H-10 · Flowering → harvest **Flowering, Fruit Set & First Harvest — PHI Required** FLOWERING → HARVEST

Gate 2 + Gate 3

■ **PHI compliance from Week 10 — organics only.** Neem / Garlic–chilli / Copper Hydroxide. No synthetic insecticide within 14 days of harvest. Reapply after >15 mm rain within 3 h of spraying (no roof protection).

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-6 to H-8	FLOWERING MANAGEMENT + NUTRITION	- High-K NPK (15-5-20 or +500 g KNO₃/500 L) every Day 6. Ca Nitrate 500 g drench twice weekly — never miss. - Spray only after 5 PM; never open flowers. Boron 1 g/L weekly through set. Tie fruit-bearing branches individually.	Flower drop: water stress, boron, spray on flowers, or >35°C. Mulch keeps soil cool.	GO High-K; Ca twice weekly; sprays after 5 PM. HOLD Spraying open flowers; missed Ca.
H-8 to H-10	FRUIT FILL + PHI + FIRST HARVEST	- Ca 500 g twice weekly to last flush; high-K continues (K drives capsaicin + colour). - From Week 10 organics only: neem alt. garlic every 3–4 days; Copper Hydroxide 2 g/L; zero synthetics. - Harvest Wk 10–12 when full-sized and colouring (orange/red/chocolate by variety). Scissors, 1 cm stem, gentle — capsaicin is an eye/skin hazard. First harvest triggers the second flush; keep feeding and pruning.	Heat (capsaicin) peaks at full ripeness — harvest early = less heat, lower value.	GO Organics only from Wk 10; each harvest logged. HOLD Synthetic within 14 d; pulling fruit off.

WEEK H-12 to H-16 · Second flush **Post-Harvest Second-Flush Management** SECOND FLUSH

Gate 2 + Gate 4

▸ **Second and third flushes out-earn the first** — establishment costs are already paid. Good post-harvest canopy and nutrition can double income from the same plants. Habanero gives 3–4 flushes in PH with correct management.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
AFTER FLUSH	DRIVE THE NEXT FLUSH	- Urea top-dress 20 g/plant around the drip zone. Resume full NPK to push new growth and flowering. - Prune exhausted branches (no active buds). Top mulch back to 12 cm if thinned below 8 cm. - After each flush, inspect every crown — foot traffic compacts soil and injures roots (Phytophthora entry).	Crown rot after harvest — worker traffic damages roots at the soil line.	GO Urea in; NPK resumed; mulch topped. HOLD Crown lesion — Metalaxyl-M drench + isolate.

REPLANTING GREENHOUSE 4 & 5 — CLEAN-RESTART PROTOCOL

A between-cycle reset that breaks the previous crop's thrips, tospovirus and nematode load before a single new seedling goes in.

■ **The danger with a replant is carry-over.** Thrips, whitefly, mosaic and tospovirus survive in old plants, fallen debris, weeds and volunteer seedlings — and nematodes survive in the soil. If GH-4 and GH-5 are simply cleared and replanted, the new crop inherits the same infection on day one. This protocol is the firebreak. It is also how these two houses clear **Gate 0** and **Gate 1** before Week 0 of the bell-pepper schedule.

▲ **Timing note — you have live crops next door.** GH-1, GH-2 and GH-3 are mid-season and carrying their own thrips pressure. Do the teardown of GH-4/5 on a day you are **not** moving between houses afterwards, change or wash over-clothes before entering the live houses, and keep the GH-4/5 doors shut during their fallow break so pests don't simply walk next door.

Step 1 — Terminate and remove the old crop (Day 1)

- **Cut and bag every old plant** — stems, leaves, roots and all fallen debris. Do not shred in place, do not compost near the field. Thrips and virus live in this material.
- Carry it right off-site and burn or bury it far from all houses and the nursery. A pile at the fence re-infects the new crop.
- Lift and remove the old drip mulch/ground cover if any. Pull every weed and **every volunteer pepper seedling** inside and for 5 m around the house — these are the green bridge the virus and thrips ride across the empty period.

Step 2 — Sanitise the structure and system (Day 1–2)

- **Net walls, posts, benches, doors:** wash down with detergent, then a 10% bleach or approved disinfectant wipe/spray. Pay attention to net-wall folds where thrips shelter.
- **Drip lines:** flush with a chlorine or acid line-cleaner, then clear water; clear or replace blocked emitters. Old biofilm harbours root pathogens.
- **Tools, stakes, ties:** soak in 10% bleach 30 min or replace. Old stakes and twine carry mosaic virus.
- **Remove all old sticky traps** and dispose off-site. Fresh traps go in at Step 5, not before.

Step 3 — Gate 0: soil clearance (this is the hard checkpoint)

- **Soil solarisation:** clear polythene over both houses' beds, sealed tight, **3–4 weeks**. This is the primary nematode and pathogen kill and the low-cost core of the reset.
- **Neem cake** 250 g/hole at 2 weeks in; compost 20–30 kg/bed.
- **Take a soil + nematode sample** and send it to the lab. If nematodes or pathogens come back present → extend solarisation / biofumigant and **re-test until the report is clear**.
- Trichoderma drench after plastic is lifted to rebuild the beneficial rhizosphere.

GATE 0 — HELD

No seedling enters GH-4 or GH-5 until a **clear soil & nematode report** is on file, signed off by the agronomist. This single rule is what would have prevented RC1 — the galled-root wipeout — last season.

Step 4 — Enforce a host-free fallow (the green-bridge break)

- Keep both houses **free of any pepper or solanaceous plant** (including weeds and volunteers) for the whole fallow / solarisation window. With no host, thrips populations collapse and tospovirus has nowhere to live.

- Minimum host-free break: the 3–4 week solarisation period. Longer is better if the neighbouring houses are under heavy thrips pressure.

Step 5 — Gate 1: establishment readiness (before transplant)

- Pre-plant structure knockdown: Cypermethrin 10EC 1 ml/L over all interior surfaces within 48 h of transplant (IRAC 3A), doors shut overnight.
- Pre-plant fungicide: Mancozeb 80WP 2.5 g/L on beds/floor/posts (FRAC M3).
- **Install fresh yellow sticky traps** (1 per 6 m²) before plants go in — the early-warning net must be live from day one.
- **Pre-stock thrips inputs on-site** (Spinosad / the Punch–Vanguish–Lion Seal rotation) so a threshold-cross is actioned the same day, never delayed by procurement.
- Pressure-test the drip; peg and photograph the 30 cm spacing; confirm the scout roster covers the two new houses.

GATE 1 — SIGNED

Transplant GH-4/5 only when the **Establishment Checklist is signed**: soil clear, structure sanitised, traps up, thrips inputs on the shelf, drip and spacing verified. Then rejoin **Section A at Week 0** and run the bell-pepper schedule from the top.

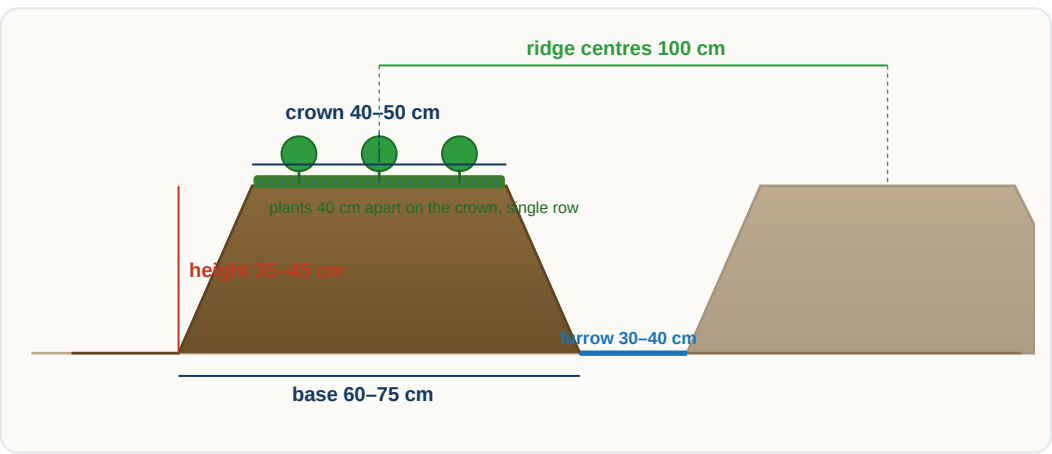
- **Once both houses are replanted**, they are simply two more zones under Gate 2 — a scouting log every day, trap counts trending, and any symptom routed through Gate 3 (diagnose before you treat). Keep the two young replants under extra watch for the first 3 weeks: they are the most vulnerable plants on the farm to thrips drifting from the older houses.

RUKPOKWU EXTENSION — OPEN-FIELD HABANERO ON RIDGES

New site establishment: land shaping, ridge geometry, spacing, drainage and the full pre-plant sequence — followed by its own complete week-by-week field schedule on the next pages.

The Rukpokwu extension is open-field habanero, so its make-or-break issue is the same one that decides every Niger-Delta field crop: **water sitting around the roots in the rainy season**. Flat beds hold water and invite Phytophthora crown rot — the disease that punishes wet feet. The whole establishment below is built to lift the plant above standing water on well-drained **ridges**, then hand over to the habanero calendar in Section B.

Ridge geometry — build to these dimensions



DIMENSION	TARGET	WHY
Ridge height	35–45 cm (use 40 cm; 45 cm on low, water-prone ground)	Lifts the crown above the wettest soil. The single biggest defence against crown rot.
Crown (flat top) breadth	40–50 cm	A stable planting surface that won't slump; room for the drip line + mulch.
Base breadth	60–75 cm	A stable batter (side slope) that survives heavy rain without collapsing.
Furrow width (between ridge bases)	30–40 cm	The drainage channel and the walking/scouting/spraying path.
Ridge spacing (centre-to-centre)	100 cm (1 m)	Crown + slopes + furrow. Standard, machine- and hand-workable.
In-row plant spacing	40 cm, single row along the crown	Open-field habanero is bushy — 40 cm gives the airflow that 30 cm can't, cutting disease and making pod-borer scouting possible.
Resulting density	~25,000 plants / ha	40 cm × 100 cm. Scales cleanly to the plot size.

Why 40 cm here, not the 30 cm used in the established field: the older field is intensively managed and pruned twice weekly to force airflow at 30 cm. On a brand-new open field, give the crop the easier margin — 40 cm in-row on a raised ridge drains and breathes far better, which is exactly what stops the crown rot and virus spread we are trying to prevent. Keep 30 cm only if you commit to the same twice-weekly sanitised pruning discipline from Section B.

Step-by-step establishment (before Habanero Week H-0)

- 1 • **Clear & level.** Remove all bush, roots and old crop debris. Rough-level the plot; note the natural fall of the land.
- 5 • **Sanitation stack.** Solarise the ridge crowns with clear polythene 3–4 weeks where feasible; everywhere apply neem cake 250 g/hole at 2 weeks before transplant. No Furadan. If the plot is too large to solarise fully, the

- **2 · Set out drainage first.** Dig a **40–50 cm deep perimeter/collector drain** around the field, running to the lowest corner and off-site. Every furrow must connect to it. In flat ground, give ridges a gentle 0.2–0.5% fall so furrows carry water away, not pond.
- **3 · Orient the ridges.** Run them so furrows drain toward the collector drain; on flat, even ground orient roughly north–south for even light. Straight, parallel ridges — not across a slope without graded furrows (erosion).
- **4 · Soil test + amend.** Sample for pH, nutrients and **nematodes** (Gate 0). Lime if pH <5.5; compost 5 t/ha (2 kg/m²) worked in along the ridge line.
- drainage + ridging + neem cake + Trichoderma + 12 cm mulch stack is your sanitation floor — but still send the nematode sample.
- **6 · Build the ridges** to the geometry above; firm the crown so it won't slump in the first storm.
- **7 · Lay drip along each crown**, one emitter per plant station at 40 cm. Pressure-test; mark and clear blocked emitters.
- **8 · Basal fertiliser:** urea 50 g per planting station, mixed into the crown.
- **9 · Mulch the crown + shoulders to 12 cm** (dry grass / rice straw / coco peat) to break Phytophthora rain-splash and hold moisture.
- **10 · Stakes 100 cm**, one per station, ready before transplant.

- **Transplant (Habanero Week H-0):** 4–6 PM only, one seedling per station over its emitter, water in immediately. Log the date as this field's **H-0, Day 1** — then run the **Rukpokwu week-by-week field schedule** that follows (microbial inoculant, IPM scouting, twice-weekly sanitised pruning, pod-borer monitoring, furrow/drain maintenance, PHI organics from Week 10, and the second-flush cycle). The ridges and drainage you built here are what let that schedule succeed on open ground.

RUKPOKWU EXTENSION — WEEK-BY-WEEK FIELD SCHEDULE

Open-field habanero on ridges · single row at 40 cm · full sun. Run this from transplant (H-0) straight through to the second flush. It carries the same habanero agronomy as the established field, with the drainage and ridge discipline an open site needs.

WEEK H-0 · Transplant on ridges Final Readiness, Transplant & First Days ESTABLISHMENT Gates 0 + 1

▲ **Ridges and drainage are already built (see the establishment section).** This week is about clearing the last gate and getting seedlings onto the crowns. On open ground the whole crop lives or dies by drainage — a furrow that ponds after rain is the first thing to fix, every time. Plant a single row at **40 cm** along each crown.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
3–4 WKS BEFORE	GATE 0 — SOIL CLEARANCE	- Ridge crowns solarised with clear polythene 3–4 weeks where feasible; neem cake 250 g/station at 2 weeks before; compost worked into the ridge line. - Soil + nematode sample sent and returned CLEAR, signed off by the agronomist. If positive → remediate and re-test. - No seedling touches a ridge until that report is clear — this is the rule that would have prevented the Season-1 galled-root loss.	A missing or unclear nematode result is a hard stop, not a delay.	GO Clear soil report on file; neem + compost in. HOLD No/positive report — remediate and re-test.
Day 1–3	GATE 1 — FINAL RIDGE, DRIP & DRAIN CHECK	- Ridges firm to spec: 35–45 cm high, 40–50 cm crown. Re-firm any that slumped. - Drip pressure-tested along each crown, one emitter per station at 40 cm; mark and clear blocked emitters. 12 cm mulch on crown + shoulders. Perimeter/collector drain (40–50 cm) dug to the outlet; every furrow connects to it and runs clear. - Stakes 100 cm staged; fresh sticky traps and thrips inputs on-site; scout roster covers the extension. Sign the Establishment Checklist.	Walk the plot and find any low spot that will pond — fix the fall before planting, not after.	GO Ridges firm; drip tested; drains clear; checklist signed. HOLD Ponding furrow or untested drip — fix first.
Day 5 · 4–6 PM	TRANSPLANT HABANERO ON THE CROWN	- Transplant 4–6 PM only. One seedling per station on the crown, directly over its emitter, single row at 40 cm. - Firm gently — no air pockets. Water each row as you plant. - Drive the 100 cm stake beside each plant now. Logbook: today = Rukpokwu WEEK H-0, Day 1.	Seedlings hardened 7+ days; crown soil moist, not dusty; not planted in mid-day sun.	GO 4–6 PM; hardened; crown moist; stakes in. HOLD Mid-day heat or unhardened seedlings.
Day 6–7	WATERING + DRAINAGE CHECK	- Field irrigation, all weeks: 6:30 AM 30 min (dry) / 20 min (rainy); 5 PM 20 min (skip if >15 mm rain today). After >25 mm rain, skip the next morning entirely. - After every rain, walk the furrows and perimeter drain — standing water = blockage; clear it the same day. - Flag any plant not upright. Day 1–3 wilt is normal; not recovered by Day 3 evening → read the roots.	Water sitting in a furrow around the ridge base is exactly what starts crown rot.	GO Drip on schedule; furrows draining; plants recovering. HOLD Ponding furrows or full drip after >25 mm rain.

WEEK H-1 to H-2 · Establishment Root Establishment — Inoculant, Ca+K, IPM, First Pruning ESTABLISHMENT

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-1 · Day 1	MICROBIAL INOCULANT (500 L field tank)	- Trichoderma 2.5 g/L (mix in 25 L, into 500 L tank, drip 30 min) — every 14 days. Or Bacillus subtilis / Azospirillum 5 ml/L. - Open-field soil holds Trichoderma well, building a lasting shield against Phytophthora/Fusarium/nematodes around the crown. - Keep 48 h clear of any fungicide drench.	48 h gap is stricter in field — fungicide persists longer in soil.	GO Biweekly; 48 h clear; soil moist. HOLD Same day as fungicide.
H-1 · Day 3	Ca + K FOLIAR (FIELD)	- Ca Nitrate 50 g + K Nitrate 50 g / 16 L, 4–6 PM, both leaf surfaces (habanero leaves are waxy). - Reapply next dry evening if rain within 2 h. Continue foliar through Week 3, then switch to drench.	Check the sky — sudden PH showers Jun–Oct wash it off.	GO Clear 2 h after spray; 4–6 PM. HOLD Rain within 2 h.
H-1 · Day 6	INSECTICIDE (IRAC 4A) + NPK 20-10-10	- Thiamethoxam (IRAC 4A) 0.2 g/L, 4–7 PM. From Week 6 add Chlorantraniliprole (IRAC 28) for pod borer. - NPK 20-10-10: 1 kg in 500 L field tank → drip 25 min. Log IRAC 4A. - Scout the ridge ends and windward edge first — pests enter an open field from the perimeter.	Field pest pressure (thrips, aphids, borer) is higher than the houses — perimeter first.	GO IRAC 4A rotated; edges scouted. HOLD —
H-2 · Day 3	PRUNING #1 (30 cm protocol, 40 cm easier)	- Sanitise the blade between every plant. Remove laterals below the Y-fork and any leaf touching a neighbour. Confirm the air gap. Copper 2 g/L on wounds. - Twice weekly (Day 3 + Day 6) from here. At 40 cm the canopy stays open more easily than the 30 cm field — keep it that way.	Habanero is bushy; even at 40 cm it will close if you skip a prune.	GO Sanitised; air gap held; wounds treated. HOLD Skipping a field prune.
H-2 · Day 6	FUNGICIDE (M3) + NPK + Mg + CROWN CHECK	- Mancozeb (FRAC M3) 2.5 g/L on dry leaves only (needs 2 dry hours to bind); delay if rain. - NPK 15-15-15 1 kg / 500 L; Mg Sulphate 500 g separate pass. Log FRAC M3. - Check every crown at the soil line — the ridge protects it, but confirm no dark Phytophthora lesion.	Mancozeb on wet leaves won't adhere; a crown lesion is an early rot warning.	GO Dry leaves; crowns clean; FRAC M3 logged. HOLD Rain/wet leaves — wait. Lesion — Metalaxyl-M drench.

WEEK H-3 to H-5 · Vegetative Vegetative Frame — Stakes, Boron, Pruning, Pod-Borer Watch VEGETATIVE

Gate 2 live

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-3 · Day 1	STAKING + Ca+K DRENCH SWITCH + BORON	- Confirm 100 cm stakes on every plant (habanero gets top-heavy). Switch Ca+K to drench: Ca Nitrate 500 g → 500 L → 20 min → flush 5 min; K Nitrate 500 g SEPARATE pass. - Boron foliar begins 1 g/L on tips, weekly through fruit set (hollow-pod prevention).	Ca and K mixed in one pass block emitters and cancel out.	GO Stakes in; Ca/K separate; Boron 1 g/L. HOLD Ca + K in one pass.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-3 – H-5 · Day 3 & 6	TWICE-WEEKLY PRUNING + SPRAY ROTATION	- Prune twice weekly, sanitised, keep the air gap, re-tie after each prune. - H-3 Day 3 garlic / Day 6 NPK + Copper Oxy (M1) + insecticide rotate. H-4 Day 3 neem / Day 6 Ca+K + NPK + Metalaxyl (4+M3) + Spinosad (5). H-5 Day 3 garlic / Day 6 high-K + Boron + Mancozeb (M3). - Zinc 2 g/L monthly (first now).	Crown lesion at the soil line → Metalaxyl-M drench that plant + its 4 neighbours.	GO Twice-weekly prune + rotation logged. HOLD Crown-rot lesion — drench now.
H-3 – H-5 · Day 2	POD-BORER MONITORING	- Inspect perimeter plants every week from Week 3 for Helicoverpa: cream/brown caterpillars boring buds/stems, entry holes with dark frass. - Threshold: any 3 plants with active entry → Chlorantraniliprole (IRAC 28) 0.5 ml/L, 4–7 PM. Log 'IRAC 28 — pod-borer threshold'.	Damage inside a pod is invisible until you cut it — scout weekly, act early.	GO Perimeter checked; act at 3 plants. HOLD >10 plants with entry holes — spray whole field today.
H-3 – H-5 · Day 3 & 6	FURROW, DRAIN & RIDGE MAINTENANCE (after every heavy rain)	- Walk all furrows and the perimeter/collector drain; clear silt and debris so water exits to the outlet. - Re-firm any ridge shoulder the rain has eroded; top the crown mulch back to 12 cm wherever it has thinned below 8 cm. - This is the open-field equivalent of the greenhouse pruning discipline — never skip it in the rains.	A silted drain turns your whole extension into a Phytophthora bath in one wet week.	GO Drains clear; shoulders firm; mulch at 12 cm. HOLD Standing water anywhere — clear before the next rain.

WEEK H-6 to H-10 · Flowering → harvest Flowering, Fruit Set & First Harvest — PHI from Week 10 FLOWERING → HARVEST

Gate 2 + Gate 3

■ PHI compliance from Week 10 — organics only. Neem / Garlic–chilli / Copper Hydroxide; no synthetic insecticide within 14 days of harvest. With no roof, reapply after >15 mm rain within 3 h of any spray.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
H-6 to H-8	FLOWERING MANAGEMENT + NUTRITION	- High-K NPK (15-5-20 or +500 g KNO3/500 L) every Day 6. Ca Nitrate 500 g drench twice weekly — never miss. - Spray only after 5 PM; never open flowers. Boron 1 g/L weekly through set. Tie each fruit-bearing branch (clusters are heavy).	Flower drop: water stress, boron short, spray on flowers, or >35°C. Mulch keeps the crown cool.	GO High-K; Ca twice weekly; sprays after 5 PM. HOLD Spraying open flowers; a missed Ca drench.
H-8 to H-10	FRUIT FILL + PHI + FIRST HARVEST	- Ca 500 g twice weekly to the last flush; high-K continues (K drives capsaicin + colour). - From Week 10 organics only: neem alternating garlic every 3–4 days; Copper Hydroxide 2 g/L; zero synthetics. - Harvest Wk 10–12 when full-sized and colouring (orange/red/chocolate by variety). Scissors, 1 cm stem, gentle — capsaicin is an eye/skin hazard. First harvest triggers the second flush.	Heat (capsaicin) peaks at full ripeness — harvest early = less heat, lower value.	GO Organics only from Wk 10; each harvest logged. HOLD Synthetic within 14 d; pulling fruit off the plant.

WEEK H-12 to H-16 · Second flush Post-Harvest Second & Third Flush SECOND FLUSH

Gate 2 + Gate 4

› **The second and third flushes out-earn the first** — establishment is already paid for. Good post-harvest canopy, nutrition and drainage can double income from the same ridges. Habanero gives 3–4 flushes in PH with correct management.

DAY	TASK	WHAT TO DO — STEP BY STEP	WATCH FOR	GO / HOLD
AFTER FLUSH	DRIVE THE NEXT FLUSH + REBUILD THE RIDGES	• Urea top-dress 20 g/plant around the drip zone; resume full NPK to push new growth and flowering. • Prune exhausted branches (no active buds). Top crown mulch back to 12 cm. • Rebuild any ridge shoulders slumped by the rains; re-cut furrows and clear the perimeter drain before the next wet spell. • Inspect every crown after harvest — worker foot traffic compacts soil and injures roots (Phytophthora entry).	Crown rot after harvest — foot traffic and a tired ridge are the two openings.	GO Urea in; NPK resumed; ridges rebuilt; mulch at 12 cm. HOLD Crown lesion — Metalaxyl-M drench + isolate.

PEPPER PEST & DISEASE DIAGNOSIS KIT

Diagnose before you treat (Gate 3). Match the symptom, confirm it, then act — and log every case with a photo.

The rule: no treatment is applied until the cause is named. Guessing is what lost Season 1 — the team sprayed the visible thrips while a virus and galled roots did the killing. Use the **quick triage** below to narrow it down in the field, confirm with the stated test, then read the matching card for cause, prevention, detection and treatment. If you cannot confirm it, photograph it and route it to the agronomist/lab — that is a diagnostic request, not a delay.

Quick field triage — what you see → what it likely is → confirm → first action

WHAT YOU SEE	LIKELY CAUSE	QUICK CONFIRM	FIRST ACTION
Silvery flecks / streaks on leaves & growing tips; tiny slivers that jump	Thrips (also tospovirus risk)	Tap a tip over white paper — thin slivers move; check trap count	Rotate thrips spray (Spinosad IRAC 5) if over threshold; log & re-count. Diagnose before spraying.
Leaves curl down, tips distort, bronze/greasy sheen, no insect visible	Broad mite	Inspect youngest tips with a 10× loupe — glassy oval mites/eggs	Abamectin or wettable sulphur on tips; remove worst tips; treat before it spreads.
Fine webbing under leaves; pale stippling; leaves dry out	Spider mite	Look under leaves for webbing + moving dots	Miticide (rotate); raise humidity slightly; remove worst leaves.
Sticky honeydew + black sooty mould; tiny white flies rise when disturbed	Whitefly (virus vector)	Trap count >5/trap or >2/leaf	Thiamethoxam (IRAC 4A) / rotate; yellow traps; sanitation.
Curled new leaves, clusters of soft green/black insects on shoots	Aphids (virus vector)	Colony on >5% of plants	Spray + rotate; remove heavily infested tips.
Ring/line patterns, mottling, bronzing; stunted, one-sided growth	Tospovirus (TSWV) — thrips-borne	Pattern + thrips history; lab confirm if unsure	No cure — rogue & bag infected plants; control thrips hard; protect the rest.
Yellow-green mosaic/mottle on leaves; puckered, distorted growth	Mosaic virus (tool/hand-borne)	Pattern; spreads along a row after pruning	Rogue & bag; sanitise blade between every plant; wash hands; control aphids.
Sudden wilt of whole plant; brown/black lesion at soil-line crown; roots rot	Phytophthora crown/root rot	Cut the crown — brown streak inside; wet soil history	Improve drainage; Metalaxyl-M drench plant + 4 neighbours; remove dead plants.
Plant wilts by day, recovers at night, then dies; one-sided yellowing	Fusarium wilt	Cut lower stem — brown streaks in the vascular ring	No cure — remove; rotate; solarise; use clean soil/beds next cycle.
Rapid whole-plant wilt, green (not yellow) leaves, stem base oozes	Bacterial wilt (Ralstonia)	Cut stem in clear water — milky bacterial ooze streams out	No cure — remove & bag; avoid water spread; solarise; do not replant peppers there.
Dark, water-soaked, scabby spots on leaves & fruit	Bacterial spot	Spots have a yellow halo; worse after rain	Copper sprays; remove worst leaves; avoid working plants when wet.
Sunken, dark, concentric spots on fruit; salmon spore ooze	Anthracnose (fruit)	Spots enlarge with concentric rings	Remove & destroy infected fruit; rotate fungicide; tighten harvest interval.

WHAT YOU SEE	LIKELY CAUSE	QUICK CONFIRM	FIRST ACTION
Grey fuzzy mould on stems/fruit in humid corners	Botrytis (grey mould)	Grey spores on dead/wounded tissue	Improve airflow (prune); remove affected parts; rotate fungicide.
White powdery patches on leaf undersides; yellow blotches on top	Powdery mildew	White powder wipes off	Sulphur/rotate fungicide; airflow; remove worst leaves.
Seedlings topple at soil line; thin, water-soaked stem	Damping-off (nursery)	Stem pinched & brown at the base	Reduce watering; Trichoderma; clean media; discard collapsed seedlings.
Dark, sunken, leathery patch at the fruit's bottom end	Blossom-End Rot (calcium)	Position (blossom end) + Ca-drench history	Raise Ca drench to 1.8 kg/1,000 L; regularise irrigation. Not a disease.
Hollow, light, mis-shapen pods; poor fruit set, flower drop	Boron deficiency	Endemic PH soil issue; check boron program	Borax 1 g/L weekly on tips (never >1.5 g/L). Not a pest.
Pale, sunken, papery patch on the sun-facing side of fruit	Sunscald	Exposed fruit after heavy leaf strip	Keep some canopy shading fruit; don't over-prune fruit-shading leaves.
Ring or radial cracks on ripening fruit	Fruit cracking	Followed a dry spell then heavy water/rain	Keep irrigation regular; steady Ca; harvest promptly in wet spells.
Chewed pods, entry holes with dark frass; caterpillar inside	Helicoverpa pod borer	Cut a suspect pod — larva + frass inside	Chlorantraniliprole (IRAC 28) at threshold (3 plants); scout perimeter weekly.
Roots knotted with galls; stunted, wilting, nutrient-starved plants	Root-knot nematode	Pull a plant — beaded galls on the roots	No rescue for galled roots — Gate 0 soil test + solarisation next cycle; rogue severe plants.
Seedlings/young stems cut off at soil line overnight	Cutworm	Grey curled caterpillar just under the soil nearby	Neem cake at planting; hand-collect at dusk; collar barriers.

Diagnosis cards — cause · prevention · detection · treatment

PESTS

Thrips

CAUSE

Tiny slender insects that rasp leaf and flower tissue. The critical threat is not the feeding — it is that thrips carry **tomato spotted wilt virus (TSWV)**. This is the RC2 killer from Season 1.

DETECTION

Daily trap counts vs the agronomist's threshold (>10/trap or >3/growing tip). Tap growing tips over white paper — slivers move. A rising trend is the early warning.

PREVENTION

Sticky traps from day one (1/6 m²); prophylactic IPM; thrips inputs pre-stocked on-site (Gate 1); weeds/volunteers removed (their off-crop hosts).

TREATMENT

Threshold crossed → same-day rotation spray (Spinosad IRAC 5, rotating with the Punch / Vanguish / Lion Seal program), after 4 PM; re-count to confirm knockdown. Diagnose before you spray.

PESTS

Broad mite

CAUSE

Microscopic mites that attack the youngest growing tips; a known suspect in the current habanero canopy problem. Invisible to the naked eye — damage looks like a virus or herbicide injury.

DETECTION

Down-curved, distorted, bronzed/greasy tips with no visible insect. Confirm with a loupe: glassy oval mites and eggs on the underside of the youngest leaves.

PREVENTION

Scout youngest tips with a 10× loupe; avoid moving mites on hands/tools between plants; keep plants unstressed.

TREATMENT

Abamectin or wettable sulphur directed at the tips (rotate, observe PHI); remove and bag the worst tips. Treat early — it moves fast through a young canopy.

PESTS Whitefly**CAUSE**

Small white flies that suck sap, excrete honeydew (→ sooty mould) and transmit begomoviruses.

DETECTION

Flies rise in a cloud when foliage is disturbed; trap count >5/trap or >2/leaf; sticky honeydew and black mould.

PREVENTION

Yellow sticky traps; net-wall vigilance; remove heavily infested lower leaves; weed control.

TREATMENT

Thiamethoxam (IRAC 4A) or rotate to the next group; spray undersides 4–7 PM; never repeat an IRAC group.

PESTS Spider mite**CAUSE**

Tiny mites under leaves in hot, dry, dusty conditions; fine webbing; pale stippling.

DETECTION

Fine webbing and moving specks under leaves; upper surface stippled pale then bronzed.

PREVENTION

Avoid dust and drought stress; keep humidity from crashing; scout undersides in dry spells.

TREATMENT

Miticide (rotate actives); remove worst leaves; correct the drought/dust stress driving it.

PESTS Cutworm**CAUSE**

Night-feeding caterpillars that sever young seedlings at the soil line.

DETECTION

Seedlings cut off at ground level overnight; a grey curled larva in the soil close by.

PREVENTION

Neem cake in planting holes; clean, weed-free beds; collars around transplants.

TREATMENT

Hand-collect at dusk; neem cake; targeted bait only if severe.

PESTS Aphids**CAUSE**

Soft green/black insects clustering on shoots and undersides; sap-feeders and efficient virus vectors.

DETECTION

Curled new growth, sticky honeydew, visible colonies; threshold = colony on >5% of plants.

PREVENTION

Scouting; encourage/keep natural enemies; remove infested tips; control ants that farm them.

TREATMENT

Spray and rotate IRAC; neem/soap for light pressure; cut out heavily infested tips.

PESTS Helicoverpa pod borer**CAUSE**

Caterpillar (*Helicoverpa armigera*) that bores into buds and pods; endemic in PH open fields. Can take 40–60% of a habanero crop.

DETECTION

Cream/brown caterpillars, entry holes with dark frass; internal pod damage is invisible until you cut a pod.

PREVENTION

Weekly Day 2 scouting of perimeter plants from Week 3; pheromone/light monitoring; Bt for young larvae.

TREATMENT

At threshold (any 3 plants with active entry) → Chlorantraniliprole (IRAC 28) 0.5 ml/L, 4–7 PM; whole-field spray if >10 plants hit.

SOIL Root-knot nematode (*Meloidogyne*)**CAUSE**

Soil roundworms that gall the roots so the plant cannot take up water or nutrients. This is RC1 — the Season 1 wipeout.

DETECTION

Stunting, mid-day wilting and nutrient-starvation that no spray fixes. Confirm by pulling a plant — beaded galls along the roots (mid-cycle root checks catch escapees).

PREVENTION

Gate 0: pre-plant soil + nematode assay, cleared before planting; solarisation 3–4 wks; neem cake; crop rotation; never replant into an untested bed.

TREATMENT

There is no rescue for galled roots. Isolate the sub-zone, apply the agronomist's nematode protocol within PHI limits, and if severe terminate + solarise that sub-zone. Fix it at Gate 0 next cycle.

DISEASE **Phytophthora blight / crown & root rot****CAUSE**

Oomycete (*P. capsici*) thriving in wet, poorly drained soil; rots crowns and roots and causes sudden collapse. The reason Rukpokwu is on ridges.

DETECTION

Sudden wilt; dark lesion at the soil-line crown; brown streak inside a cut crown; clusters of death in the wettest spots.

PREVENTION

Drainage & ridging; 12 cm mulch to stop rain-splash; avoid waterlogging; Trichoderma; copper in the rotation.

TREATMENT

Improve drainage immediately; Metalaxyl-M drench the plant + its 4 neighbours (rotation only); remove dead plants; never let it sit in standing water.

DISEASE **Bacterial wilt (*Ralstonia*)****CAUSE**

Aggressive soil bacterium causing rapid wilt with leaves still green; spreads in water and soil.

DETECTION

Fast whole-plant wilt, green leaves; the giveaway test — cut the stem base in a glass of clear water and milky bacterial threads stream out.

PREVENTION

Clean soil, good drainage, rotation, sanitation of tools; avoid water movement from infected areas.

TREATMENT

No cure — remove and bag; stop water spread; solarise; do not replant solanaceous crops there.

DISEASE **Botrytis (grey mould)****CAUSE**

Fungus on wounded or senescing tissue in humid, still air; grey fuzzy sporulation.

DETECTION

Grey fuzzy mould on stems, flowers or fruit, especially in the moist zone under large fruit.

PREVENTION

Airflow (the 5 cm gap + door-flush ventilation); remove dead tissue and pruning debris; don't leave wounds wet.

TREATMENT

Improve airflow; remove affected parts; rotate fungicide; avoid dense wet canopy.

DISEASE **Damping-off (nursery)****CAUSE**

Soil fungi collapsing seedlings at the soil line in wet, crowded nurseries.

DETECTION

Seedlings topple; thin, water-soaked, pinched brown stem at the base.

PREVENTION

Clean media and trays; don't over-water; good airflow; Trichoderma-treated media.

TREATMENT

Cut watering back; improve airflow; discard collapsed seedlings; drench media with Trichoderma.

DISEASE **Fusarium wilt****CAUSE**

Soil fungus blocking the plant's water vessels; slow one-sided wilt and death; persists in soil for years.

DETECTION

Day wilt with night recovery, then death; brown streaks in the vascular ring of a cut lower stem.

PREVENTION

Clean/tested soil, solarisation, rotation, resistant varieties where available; avoid moving infested soil.

TREATMENT

No cure — remove affected plants; do not replant peppers into that soil without solarisation/rotation.

DISEASE **Anthracnose (fruit)****CAUSE**

Fungus (*Colletotrichum*) causing sunken, rotting fruit spots — a major market loss on ripening pods.

DETECTION

Sunken dark spots with concentric rings and salmon-coloured spore ooze on fruit.

PREVENTION

Airflow via pruning; fungicide rotation; remove infected fruit fast; avoid overhead wetting.

TREATMENT

Remove and destroy infected fruit; rotate to an effective fungicide; tighten interval; harvest promptly.

DISEASE **Powdery mildew****CAUSE**

Fungus giving white powder on leaves; reduces photosynthesis and yield.

DETECTION

White powdery patches (undersides first) with yellow blotches on the upper surface; powder wipes off.

PREVENTION

Airflow; avoid dense canopy; scout undersides; sulphur in the program.

TREATMENT

Sulphur or rotate fungicide; remove worst leaves; open up the canopy.

VIRUS **Tospovirus (TSWV)****CAUSE**

Thrips-borne virus — the RC2 damage. Once systemic it cannot be cured; the only lever is controlling the thrips vector early.

DETECTION

Ring/line patterns, bronzing, one-sided stunting; correlate with thrips history; lab-confirm if unsure.

PREVENTION

Control thrips hard from day one (Gate 1/2); remove weed hosts and volunteers; rogue infected plants early.

TREATMENT

No cure — rogue and bag infected plants, and hit the thrips population hard to protect the rest. This is why RC2 control is same-day.

VIRUS **Mosaic viruses (tool/contact-borne)**

CAUSE Sap-transmissible viruses spread mainly on unsanitised blades and hands during pruning — they can travel down a whole row in a week.	PREVENTION Sanitise the blade between every plant; wash hands; rogue infected plants; control aphids; avoid tobacco contact.
DETECTION Yellow-green mosaic/mottle, puckered distorted leaves; a line of affected plants following the pruning direction.	TREATMENT No cure — rogue and bag; reinforce tool sanitation; remove the source plants before they seed a row.

DISORDER **Boron deficiency / hollow pod**

CAUSE Endemic Niger-Delta soil deficiency causing hollow, mis-shapen pods, poor set and flower drop.	PREVENTION Weekly boron (Borax 1 g/L) on tips through fruit set; keep 4 h apart from Ca.
DETECTION Hollow, light, deformed pods; poor fruit set; flower drop despite good pest control.	TREATMENT Borax exactly 1 g/L weekly on tips — never exceed 1.5 g/L (scorch). Weigh, don't eyeball.

DISORDER **Blossom-End Rot (Ca)**

CAUSE A calcium disorder (not a disease) from a Ca gap or irregular water during fruit fill.	PREVENTION Never miss a Ca drench; keep irrigation regular; split Ca dosing (Day 1/Day 4).
DETECTION Dark, sunken, leathery patch at the blossom (bottom) end of the fruit.	TREATMENT Raise Ca Nitrate drench to 1.8 kg/1,000 L; regularise watering. Affected fruit is lost but the rest are saved.

DISORDER **Sunscald & fruit cracking**

CAUSE Sunscald: exposed fruit burns on the sun side. Cracking: erratic water (dry spell then flood) splits ripening fruit.	PREVENTION Keep some canopy shading fruit (don't over-strip); keep irrigation steady; steady Ca.
DETECTION Sunscald — pale papery patch on the sun-facing side. Cracking — ring/radial splits on ripening fruit.	TREATMENT Restore fruit-shading canopy; regularise irrigation; harvest promptly during wet spells.

▲ **Every treatment follows a diagnosis, and every diagnosis and spray is logged** — that record is what lets the owner verify from offshore and what turns this season's problem into next season's prevention (Gate 4).

QUICK-REFERENCE TABLES

Rotations, mixing rules, spray prep, irrigation, deficiency, EC/pH and grading — the whole reference bench in one place.

Insecticide rotation — IRAC groups (never the same group twice in a row)

ORDER	PRODUCT	IRAC	RATE	SPRAY TIME	TARGETS
1st	Cypermethrin 10EC	3A pyrethroid	1 ml/L	4–7 PM	General insects
2nd	Thiamethoxam 25WG	4A neonicotinoid	0.2 g/L	4–7 PM	Whitefly, aphids
3rd	Spinosad 45SC	5 spinosyn	0.3 ml/L	After 4 PM only	Thrips (best)
4th	Chlorantraniliprole 200SC	28 diamide	0.5 ml/L	4–7 PM	Caterpillars, borer, thrips
5th	Restart at 3A	—	—	—	Restart rotation

Fungicide rotation — FRAC groups (never the same group twice in a row)

ORDER	PRODUCT	FRAC	RATE	INTERVAL	CONTROLS
1st	Mancozeb 80WP	M3 contact	2.5 g/L	10–14 d	Broad-spectrum fungal
2nd	Copper Oxychloride 50WP	M1 contact	3 g/L	10–14 d	Phytophthora, bacterial
3rd	Metalaxyl-M + Mancozeb	4+M3 systemic+contact	2.5 g/L	10–14 d	Phytophthora (systemic)
4th	Restart at M3	—	—	—	Restart rotation

Mixing rules — never combine

COMBINATION	CORRECT ACTION
Ca Nitrate + K Nitrate (same tank)	Separate passes; flush between.
Ca Nitrate + Mg Sulphate (same spray)	Different days / separate passes.
Ca Nitrate + Phosphate	Separate passes; flush between.
Neem oil without soap	Always premix 150 ml oil + 30 ml soap.
Borax above 1.5 g/L	Never exceed 1.5 g/L — scorch.

Spray preparation & PHI

PRODUCT	PREP	PHI
Garlic–Chilli	6 g + 6 g steeped 6 h in 4 L, strain twice, +15 ml soap +15 ml vinegar → 15 L. Same day.	0 d
Neem oil	Cold-pressed 150 ml + 30 ml soap, emulsify, → 16 L. Use within 8 h.	0 d
Trichoderma (GH)	5 g/L, bucket first, → 1,000 L, drip 30 min. Every 14 d.	0 d
Copper Hydroxide	50WP 2 g/L, after 5 PM. PHI-safe.	0 d

COMBINATION	CORRECT ACTION
Metalaxyl-M consecutively	Only in FRAC rotation, max every 3rd.
Foliar Ca after first open flower	Drench-only Ca from first flower.

Irrigation quick reference

CROP	SESSION	SKIP IF
GH bell	6:30 AM 20 min	Never
GH bell	1 PM 15 min	Net-wall condensation + moist 5 cm
GH bell	5 PM 20 min	Never
Field haba	6:30 AM 30/20 min	>25 mm rain previous day
Field haba	5 PM 20 min	>15 mm rain today

EC & pH targets

MEASURE	TARGET
pH irrigation water	5.8–6.5 (if >7.0 add citric acid 5 g/10 L)
EC plain water	0.3–0.8 mS/cm
EC fertigation (veg)	1.8–2.8 mS/cm
EC fertigation (fruit)	2.0–3.0 mS/cm
EC >3.5 mS/cm	Stop — flush 60 min before fertigating

Nutrient deficiency — quick diagnosis

SYMPTOM	MISSING	ACTION
New tip leaves pale yellow	Nitrogen	+20% NPK; check drip reaches all plants
Lower leaves yellow, veins green	Magnesium	MgSO4 50 g/16 L foliar, weekly
Dark sunken spot at fruit base	Calcium + water	Ca Nitrate 1.8 kg/1,000 L; steady water
Hollow / deformed pods	Boron	Borax 1 g/L weekly (max 1.5)
Pale fruit colour	Potassium	NPK 15-5-20 or +500 g KNO3/tank
Interveinal yellow, middle leaves	Iron/Manganese	Check pH — >7.0 locks Fe/Mn; acidify

Harvest grading (20–30% revenue lift)

GRADE	CRITERIA	MARKET
A	Uniform, full colour, firm, no marks (bell 70–90 mm / haba 35–50 mm)	Export / hotels
B	Minor cosmetic marks; full size & colour	Local retail
C	Undersized/irregular, no rot	Processors — always sell
Reject	Rot, disease, insect through flesh, soft	Remove far from crop